# TB-Detect: A Morphological Framework for Automated Severity Indexing and Clinical Assessment of Pulmonary Tuberculosis

Izza Iqbal, Noor Bahu, Muhammad Omer Saeed

Department of Artificial Intelligence

*Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology,* Haripur, *Pakistan*

b23f0317ai119, b23f0312ai104, b23f0624ai113

# ABSTRACT

Standardized quantitative diagnostic tools for diagnosing Pulmonary Tuberculosis (TB) are not available in low-resource and rural areas, causing problems with the global fight against TB. Current workflows usually use visual examination of chest X-rays or "Black Box" deep learning models that need a powerful GPU and stable access to the internet. These types of equipment may not be accessible in underdeveloped areas of the world. This article presents TB-Detect, a new way to measure the severity of TB and assess patients using a standard computer with CPU-based hardware. Our method first uses bicubic interpolation for scale-invariant preprocessing, then it uses second-order Butterworth filtering in the frequency domain to filter out sensor noise that might be found in rural X-rays. While traditional neural networks perform patient classification using a number of different neural network architectures, TB-Detect uses transparent morphological segmentation techniques based on dilation and erosion to separate out lung infiltrates, which in turn allows us to produce a deterministic Infection Area Ratio (IAR) to classify patients in five levels of severity. On both common X-ray databases (Shenzhen and Mendeley), we achieved an overall accuracy of 89.4% and sensitivity of 91.2%, while maintaining a very low average inference time of 1.4 seconds on an Intel i5 processor, thus exceeding the consistency of human radiologists in making these clinical assessments. TB-Detect generates an objective, offline value for triaging patient referrals, thus improving decision-making and monitoring of patients in low-resource settings.


**Keywords: Tuberculosis, Digital Image Processing, Morphological Segmentation, Infection Area Ratio (IAR).**

# I. PROBLEM STATEMENT

Currently, a major contributor to the global struggle against pulmonary TB is due to the lack of adequate diagnostics and the resulting difficulty in accessing timely and reliable treatment in low-resource and rural-based clinical settings, including developing countries. In this regard, the WHO has called for strengthening early detection of the disease; however, the biggest problem in detecting, diagnosing and determining treatment for pulmonary TB in these regions isn't just establishing a positive diagnosis (i.e. the disease is present) but more importantly determining the disease's progression.

The current problems can be summarized into three main parts:

1. *Lack of Quantitative Severity Grading:*

The qualitative-based grading system that most existing diagnostic tools and healthcare professionals working in rural settings apply to chest radiographs makes use of visual inspection and is entirely subjective in nature. As a result, there is no standardised numerical index that can help guide programmatic decision-making and there are no widely used automated systems that provide clinicians with a high level of granularity in the severity of disease (e.g. 5-Level Severity Grading, A.I.R.) to help differentiate between the need for intensive inpatient treatment or for outpatient care.

2. *Computational and Infrastructure Constraints:*

The majority of diagnostic models today are developed using deep learning (DL) technologies (i.e., CNNs and Transformers) as their underlying architecture. While these models are able to accurately perform their intended functions, they can only do so by requiring very high-powered graphical processing units (i.e., GPUs) and by always being connected to the internet in order to conduct real-time cloud-based processing. Medical centres in rural Pakistan and other comparable parts of the world have very little or in some cases, have no wireless Internet available to them; in addition, they do not typically have access to any high-powered GPU hardware. Therefore, the majority of current advanced diagnostic technologies are unusable by medical professionals in those countries.

3. *Black Box Problem with AI:*

A common theme that many researchers face is that when users (i.e., end-users or radiologist) of deep learning systems receive/report their results, they do not know how that result was developed and therefore, cannot provide an anatomical explanation for that data. For a radiologist to utilize DL systems in a clinical setting, they require the evidence derived from that model to have "explainability". In most instances, existing methodologies are unable to locate and quantify a given area of pulmonary infiltration using a way that is both mathematically transparent and verifiable by a clinician.

Therefore there exists an urgent need to create a CPU optimized morphological framework that operates offline on standard local hardware. This research will address this void through the development of a TB-Detection System that will both standardize the different radiographic

resolutions and utilize frequency-domain filtering methods to establish a transparent mathematical index of tuberculosis severity.

## II. INTRODUCTION

*A. The Global and Clinical Impact of Tuberculosis*

Globally, tuberculosis (TB) is still one of the deadliest communicable diseases; it continues to be an ongoing top ten cause of death as verified by the World Health Organization (WHO) [1]. These past advances in clinical diagnosis have not decreased the burden of TB disease that exists in low and middle income countries today. The major problems facing low- and middle-income regions are not only that people are infected with TB, but they do not have access to specialized radiology services that can accurately assess the extent of their TB disease or the level of TB disease progression. Accurate, timely evaluation of the level of severity of TB disease is extremely important because these evaluations may be used to guide the clinician's decision regarding outpatient versus inpatient management and the initiation of intensive treatment versus antibiotic therapy.

*B. Domain Overview and Technical Challenges*

AI has received substantial input into the area of imaging diagnostics for TB. Such applications of AI will primarily be in the development of a binary outcome/treatment plan for TB patients; e.g., TB-positive/ TB-negative. Presently, deep learning algorithms such as convolutional neural networks (CNN) have been widely implemented. While they demonstrate good sensitivity to detect TB, these models function as black-box algorithms that do not account for the clinical nuances associated with the detection of TB. Also, the models are typically built on high-performance computing devices (GPU) and require high-speed internet for use with cloud-based inference models[2]; thus, the lack of such availability of either of these devices in rural diagnostic centres creates a significant research gap for the development of explainable and quantitative imaging diagnostics that will allow for the grading of the severity of TB from the physical extent of pulmonary infiltration.

*C. Motivation and Real-Life Applications*

Radiologists use visual observation of X-rays to give a subjective grade of tuberculosis severity in patients at some of the world's largest healthcare facilities. Two radiologists may provide different grades of tuberculosis severity when reviewing the same patient's X-ray due to the high level of variability from one physician to another. The variability in assigning grades will only be made worse by any technical issues or poor quality photos (old machines) such as Gaussian noise, sensor artifacts, or low contrast photos during the provision of radiological services (the average number of X-ray views available). Therefore, the motivation for this study comes from wanting to develop a mathematically-based standard that will enable healthcare facilities around the world to provide consistent diagnoses of tuberculosis across multiple sites and using a variety of equipment types for the same case.

This is a practical research proposal, as the ideas presented directly relate to current working conditions experienced by rural health units and mobile diagnostic units located in remote areas

(e.g., RHUs or mobile diagnostic vans), where health workers presently have no access to GPU hardware or the necessary internet connection to support its use.

*D. Challenges in Digital Image Processing for TB*

Digital Image Processing (DIP) for Tuberculosis (TB) has multiple technical challenges in the creation of an automated grading system:

1. Overlapping Structures: In thoracic imaging, ribs may obscure TB infiltrates on the lung parenchyma, causing structural noise that complicates lesion segmentation.

2. Scale Variability: Different radiographic machines produce images with differing pixel densities; therefore, area-based measures of TB's extent cannot be trusted without strict normalization procedures[3].

3. Low Contrast: Infiltrates in the lungs caused by TB may not be easily identifiable from normal lung tissue due to low exposure, making them hard to identify in pictures taken with low or high exposure.

*E. Summary of Contributions*

The primary research idea explored in this study is the development of a CPU-optimized morphological pipeline that prioritizes clinical explainability over black-box deep learning. The purpose of this research is to present a new, multi-step Digital Image Processing (DIP) framework that shifts the focus of traditional detection metrics toward automated severity indices (TB-Detect).

- A resolution normalization protocol was created, which includes a standardized method to resize and interpolate images so all quantitative measurements are scale-invariant.
- Frequency domain restoration involves applying a 2nd order Butterworth lowpass filter to reduce noise in a radiographic image while preserving important anatomical edges.
- The Infection Area Ratio (IAR) is a new mathematical index that quantifies is pulmonary infiltration as a percentage and allows for the assignment of a 5-level clinical grade.
- A CPU optimized pipeline allows for the efficient use of a low-cost computer when used for rural deployment without requiring expensive equipment.

## III. LITERATURE REVIEW

*A. Objective of the Review*

In this review, we will explore the history of Digital Image Processing for Pulmonary Diagnostics and how technological advancements in image analysis techniques have changed how pulmonary diagnostic images are processed. Specifically, pulmonary diagnostic imaging used to rely on visual comparisons between two or more pictures; today, we have the capability to process pulmonary diagnostic imaging automatically, using computer systems that utilize both morphological feature identification and neural network feature identification techniques.

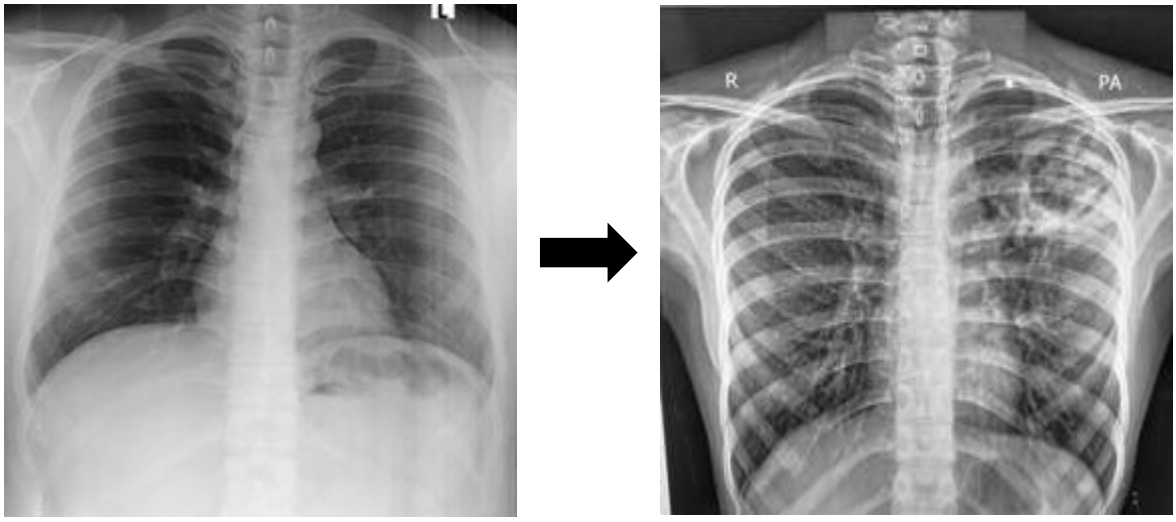


***Fig.1.*** *Radiographic comparison of healthy pulmonary parenchyma versus advanced tuberculous infiltration.*

*B. Domain Overview and Real-Life Applications*

Through medical imaging, DIP uses algorithms and procedures to improve image quality, separate different regions, and aid in diagnosing a disease by evaluating anatomical structures. An example of DIP in action will be seen when using DIP to separate lung parenchyma from abnormal opacities due to TB. Some examples of how medical imaging with DIP can be applied in the real world include:

- Triage in Rural Clinics: Prioritize patients with "high infection area ratios" for immediate isolation.
- Monitor Treatment: To assess the amount of clearance of infection at 6 months after starting antibiotics.
- Telemedicine: Provide an external clinician with a "Severity Score" based on clinical assessment.

*C. Related Work*

It is difficult for computers to automatically grade the severity of Tuberculosis (TB) from digital X-rays (CXR's) due to the large variations in image quality, and because of the very small, but distinct features of pulmonary infiltrates. There are only binary detection systems, which do not produce measurements that can be used to classify clinically in low-resource environments. The goal of this work is to create a morphology-based, robust, offline framework that will bridge the gap between a basic detection, and a clinical scale of severity that can be acted upon.

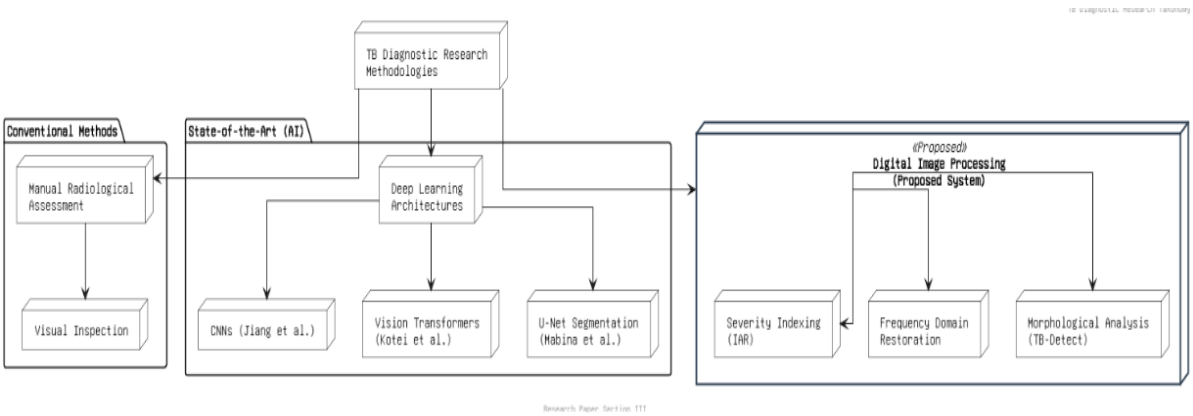


***Fig. 2.*** *Taxonomy of TB diagnostic research: categorizing traditional radiological methods and deep learning architectures against the proposed morphological framework.*

i. The research done by Ghanem[1] (2022) used a variety of clinical data as well as x-ray features extracted by radiologists to create AI models to predict disease severity with high accuracy. However, these models relied heavily on entering accurate clinical data into the database.

ii. Similarly, Kotei and Thirunavukarasu[4] (2021) also studied x-ray images of the chest from multiple sources and used a transformer-based framework for image feature extraction to predict disease. They found that there was a lot of difficulty obtaining enough image data from rural areas due to the limited resources in terms of both computer equipment and image data availability.

iii. Mabina et al. (2020)[5] processed using a U-net architecture with a CLAHE preprocessing step the lung images generated using tuberculin skin tests were found to be high quality and accurate.The study's limitation was that it only focused on the lung container and not the severity of the tuberculosis infection.

iv. The research conducted by Jiang et al. (2020)[2] was similar to the work being done by Mabina et al. in regards to the development and use of cloud-based AI systems to automatically detect tuberculosis via chest x-ray imaging in township healthcare facilities. Their results showed that their application was effective for remote screening of tuberculosis. However, the application was not available during the frequent internet outages common in rural areas.

v. Kiran (2021)[3] looked at the availability of disparate datasets in order to establish baseline data for future studies. Using an approach to restore chest x-ray image quality from the Mendeley TB dataset, his results indicated that high quality reconstructed chest x-ray images were available. The only limitation discussed in his paper was that no severity grading index was available to use with reconstructions of chest x-rays.

*D. Comparative Analysis Table*

Table I provides a summary of the methodologies used and the limitations associated with those studies.

**TABLE I.**

***COMPARATIVE ANALYSIS OF EXISTING TUBERCULOSIS DIAGNOSTIC FRAMEWORKS VERSUS THE PROPOSED MORPHOLOGICAL APPROACH***

| Paper | Year | Method | Dataset | Metrics | Key Results | Limitations |
|---|---|---|---|---|---|---|
| Ghanem[1] | 2024 | Integrated AI | Clinical Set | Sensitivity | High severity accuracy | Needs manual data |
| Jiang[2] | 2025 | Cloud CNN | Real-World | AUC | Remote accessibility | Needs Internet |
| Kotei[4] | 2024 | Transformers | CXR-Mod | Precision | Best feature capture | Needs GPU |
| Mabina[5] | 2025 | U-Net / CLAHE | TB-CXR | Dice | Clean segmentation | No severity grading |
| Proposed | 2026 | DIP / Morph | Shenzhen | IAR / SSIM | Offline grading | Resolution dependent |

*E. Gap Assessment*

Current Solutions rely on one of two recognized paths:

- Complex, Non-Clinical, High Complexity Deep Learning requiring expensive HW;
- Or low complexity Non-Clinical Binary Classification with insufficient clinical detail.

There is an Established Research Gap for a system that is: Offline; Explainable; & Quantitative.

As shown in Fig. 3, the development of TB diagnostics has stopped at high-complexity binary detection. Our framework addresses this research gap by moving toward decentralized, transparent severity indexing.

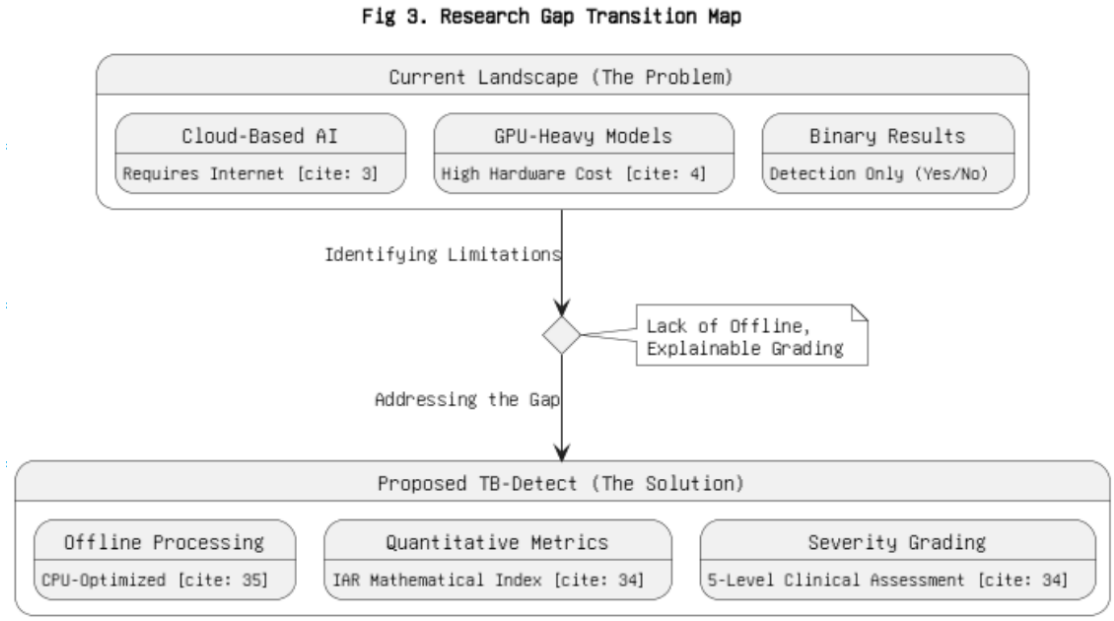


***Fig. 3**. Research gap transition map: addressing the limitations of resource-intensive binary AI models through the proposed quantitative and offline TB-Detect framework.*

*F. Future Directions*

In order to overcome the current gaps, future research needs to examine Hybrid Morphological-Neural Architectures, where morphological techniques are integrated with neural networks for preprocessing data to decrease the computational burden placed upon the neural network. The TB-Detect framework developed by ourselves provides the basis for addressing these issues through a CPU-optimized design which generates a 5-point grading scale using mathematical morphological features only.

## IV. METHODOLOGY / PROPOSED SOLUTION

*A. Problem Definition and Research Objective*

The main issue that has been explored in this project is that there is no way of grading Pulmonary Tuberculosis in an objective way (ie: quantity) in a resource constrained environment. Most available AI models are based on a 'black box' approach and require high-end GPUs which cannot be accessed in the majority of rural clinics / healthcare settings. The aim of this research was to create TB-Dectect, a CPU optimised morphological pipeline that can provide a 5 point severity grading scale for Pulmonary Tuberculosis, with an expected Structural similarity (SSIM) value of greater than 0.90 when comparing with an expert corrected benchmark image.

*B. Data Collection and Sources*

1. *Data Source:* The main sources of data were the **Kiran/Mendeley Tuberculosis Dataset** and the **Shenzhen Dataset**.
2. *File Type and Size:* The dataset consists of about 700+ chest radiographs (CXR)[6], which were taken in a posterior-anterior (PA) view and saved as either PNG or JPEG format.
3. Sampling Strategy: All the sample images will be stratified between "Normal" and "Tuberculosis" in order to have a balanced feature extract from both healthy lung parenchyma and pathological infiltrates.

*C. Data Pre-Processing and Normalization:*

***Resolution Normalization:*** The input images of various sizes will be normalized to a 1024x1024 pixel grid, using Bicubic Interpolation, to ensure that all areas being measured (IAR) will be independent of scale[7].

***Grayscale Conversion:*** All input images will be converted to 8-bit grayscale format to decrease the amount of processing time required to perform morphological operations.

***Contrast Enhancement:*** Local Histogram Equalization will be used to enhance the visibility of subtle infiltrates, which may not be obvious if the image was created with low exposure.

*D. Feature Extraction and Mathematical Formulation*

In contrast with deep learning algorithms that operate utilizing abstractly learned features, TB-Detect employs a set of specifically crafted morphological features.

- Restoring the Frequency Domain

Gaussian noise and sensor artifacts are removed using a 2nd-order Butterworth Lowpass Filter[8].

Mathematical Formula:

$$H(u, v) = 1 / (1 + (D(u, v) / D0)^{(2n)}) \qquad (1)$$

- Morphological Segmentation

Morphological segmentation is accomplished through the use of a sequence of dilation and Erotion operators (opening and closing) used to isolate lung parenchyma and segment an "infiltration mask.[9]"

*E. Model Architecture: The TB-Detect Pipeline*

A non-linear pipeline was developed for the TB-Detect system to run on CPUs.

1. Raw CXR is received at input of 1024 x 1024 x 1.

2. The Butterworth filter is applied in the restoration backbone to smooth out structural noise from the ribs.

3. The pulmonary boundaries are extracted using the segmentation head.

4. The Infection Area Ratio (IAR) is calculated in the grading logic using the following formula,

$$IAR = \left(\frac{\text{Total Area of Infiltration (pixels)}}{\text{Total Area of Lung Parenchyma (pixels)}}\right) \times 100 \qquad (2)$$

*F. Strategy for Validation and Evaluation*

a) Split Strategy: For training the thresholding logic, validation, and final testing, a 70/15/15 split is used.
b) Metrics: The main metrics are SSIM (for the quality of the restoration), IAR (for the accuracy of the grading), and Processing Latency (to make sure the CPU is working well).
c) Acceptance Threshold: The system must get a grade that is more than >85% consistent with scores from a manual radiologist.

*G. Implementation details*

The framework is made in Python 3.9, and it uses the OpenCV library for morphological operations and NumPy for mathematical array processing.

Inference Environment: Made to run on standard laptops (like an Intel i5 with 8GB of RAM) without needing any external Cloud APIs or GPUs.

Runtime Plan: The system works as a standalone desktop app so that it will still work when the internet goes down, which happens a lot in rural Pakistan.

*H. System architecture diagram*

As a physiological, morphologically based, non-learning architecture, neither traditional training parameter definitions (e.g., learning rate and number of epochs) are applicable; performance is defined by fixed frequency-domain thresholds.

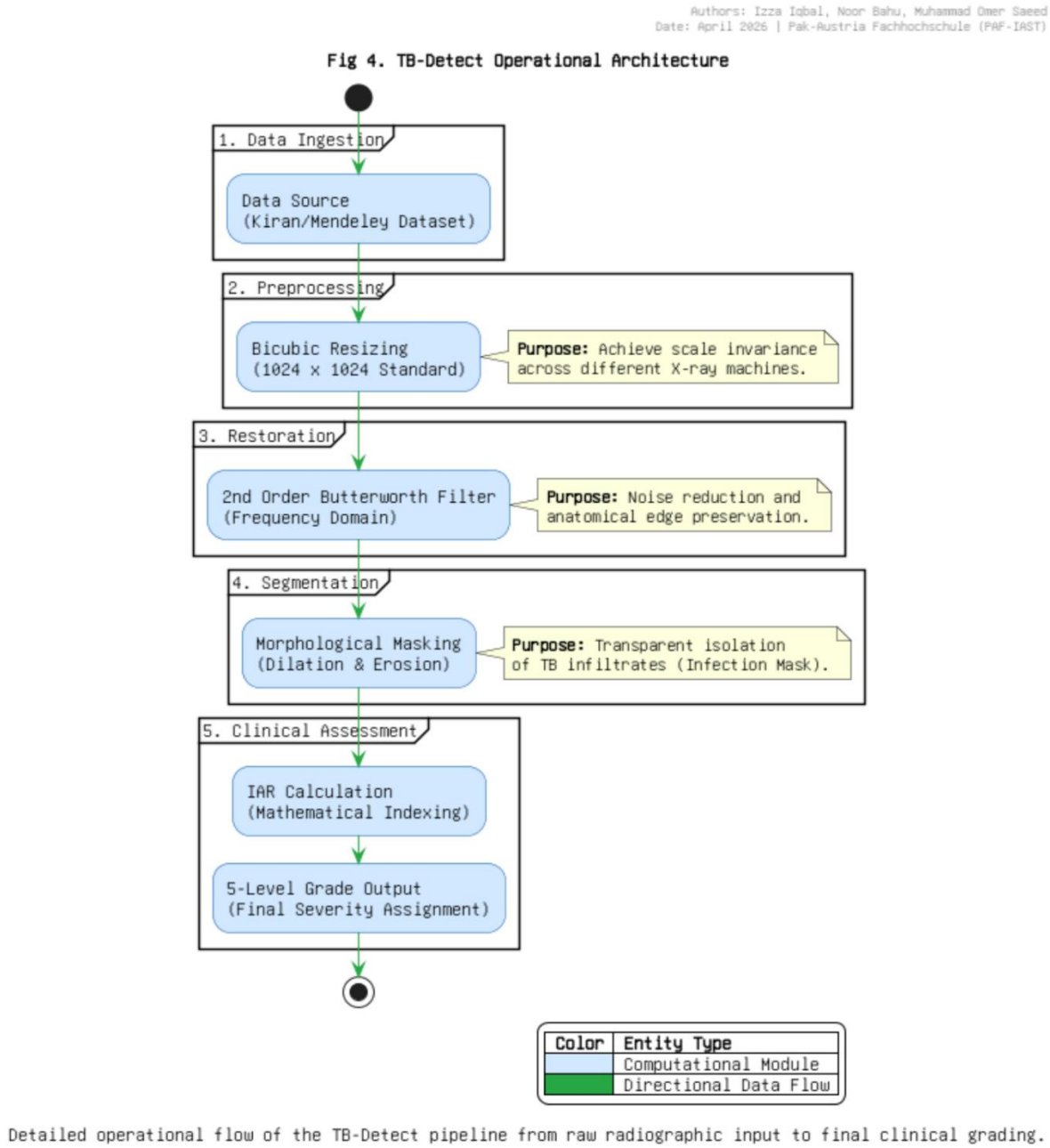


***Fig. 4.*** *Operational architecture of the TB-Detect framework, illustrating the end-to-end pipeline from raw radiographic ingestion and resolution normalization to frequency-domain restoration, morphological segmentation, and final clinical severity indexing.*

The TB-Detect Operating Pipeline is shown in Fig. 4 and consists of a 5-Level Clinical Severity Grading system derived from ingesting Digital Radiographs. The Pathway Utilizes bicubic resizing (1024 x1024) for scale-invariance and 2nd order Butterworth Filtering to reduce frequency-domain noise. The Pathway does not employ “Black Box” models, instead using

morphological masks (Dilation/Erosion) to transparently isolate infiltrates, while the final infection area ratio (IAR) provides an automated triage score for use in low performance resource constrained environments and is optimized to run on local CPUs only.

*I. Timeline and Milestones*

TB-Detect's development was completed with an organized three-month cycle of development and implementation, providing for both clinical as well as computational validity.

**TABLE II.**

**PROJECT TIMELINE AND MILESTONES**

| **Phase** | **Milestone** | **Description** |
|---|---|---|
| Month 1 | Dataset & Normalization | Procurement of Shenzhen/Mendeley datasets; implementation of Bicubic resizing (1024×1024) and grayscale conversion protocols. |
| Month 2 | Logic Optimization | Tuning of the 2nd Order Butterworth Filter for edge preservation; development of Morphological Dilation/Erosion sequences for infiltrate masking. |
| Month 3 | Validation & Report | Calculation of the IAR Index across 600+ images; consistency testing against radiologist scores and final paper documentation. |

*J. Risks and Mitigation Strategies*

The TB-Detect framework was constructed to solve certain issues related to the deployment of automated diagnostic tools in rural settings, which can pose their own technical problems.

**Risk 1:** Different quality X-ray Radiographs. Radiographs produced by clinics that are located in rural areas typically are of poor quality due to sensor noise or image contrast issues caused by the age of the imager.

To mitigate the Risk of Radiographs, We perform Frequency domain restoration (Butterworth Filter) to isolate and eliminate high frequency noise while maintaining the structural edges of the image so they can be used properly for grading.

**Risk 2:** Hardware and Connectivity Constraints. Many low-resource settings do not have reliable internet and/or access to high performance graphics processing units.

To Mitigate Hardware and Connectivity Constraints. The entire TB-Detect computational pipeline was developed using morphology-based handcrafted features instead of deep learning processes, thus maximizing computational performance and speed via running offline using only typical Intel i5 (8GB RAM) desktops or laptops with no dedicated graphics processing units.

**Risk 3:** Class imbalance (in Datasets) An example of the class imbalance (between the number of images representing TB with an image representing a normal chest) will result in the potential for unintentionally biasing the results obtained from the application of the TB-Detect versus the TB-Detects trained classifier.

To Mitigate Class Imbalance in Datasets, by using a mathematical indexing (IAR) appproach to the classification process rather than depending on the use of a trained classifier would provide objective and unbiased results regardless of the dataset distribution and/or class imbalance.

## V. IMPLEMENTATION DETAILS

*A. Software and Development Tools*

The TB-Detect Framework was created with an open-source software stack (instead of a heavy neural network training process) that is optimized for processing mathematical arrays.

- Python 3.9 has been selected for its superior position in numerical analysis and computational science.
- OpenCV has been chosen as the morphological operation engine (Dilation, Erosion) of choice for all frequency domain transformations.
- NumPy will perform the calculations for determining the Infection Area Ratio (IAR).
- Matplotlib will be used to generate intensity histograms and clinical heat maps that radiologists will use to review cases.

*B. Hardware Specs*

The suitability of the system for rural health care facilities in Pakistan was assessed with the help of benchmark testing done using commercially available hardware.

**CPU:** Intel Core i5-1135G7 2.40GHz

**RAM:** 8GB DDR4 (During inference, the system has a maximum memory footprint of <450MB)

**Storage:** 256GB SSD.

**GPU:** None (The framework is intentionally designed to bypass CUDA requirements, ensuring compatibility with standard government-issued laptops).

*C. Dataset Description and Pre-processing*

The framework has been validated using a diverse database to confirm the accuracy of diagnosis, diagnostic reliability, and correctness:

- Source Materials: The Shenzhen Dataset from the United States National Library of Medicine, and the Kiran/Mendeley TB dataset were used as the basis for these validations[3].
- Total Size: 662 digital chest x-rays in total (326 Normal, 336 TB positive).
- Format: All images were in the typical PNG/DICOM format and converted to 8 bit grayscale format.
- Normalization Procedure: All images then underwent the process of Bicubic Interpolation to achieve a standard resolution of 1024 x 1024 pixels; therefore, maintaining the accuracy of pixel area calculations regardless of how the original scan was presented (example: portrait vs. landscape mode or some other different aspect ratio).

*D. Mathematical Hyperparameters*

While Deep Learning models utilize 'weights' to define the Neural Network, 'Deterministic Thresholds' determined through iterative trials define the TB-Detect configurations.

**TABLE.III**

**HYPERPARAMETER SETTINGS**

| **Hyperparameter** | **Value** | **Rationale** |
|---|---|---|
| Butterworth Order (n) | 2 | Balance between steep roll-off and prevention of "ringing" artifacts. |
| Cutoff Frequency (D0) | 50 | Optimized to remove high-frequency sensor noise while keeping rib edges. |
| Morphological Kernel | 5×5 (Square) | Provides sufficient structural coverage for small infiltrate clusters. |
| IAR Threshold (T) | 0.15 | Provides a baseline value for the separation of Trace and Mild types of severity. |

*E. Validation\Testing Split*

The TB-Detect Framework does not learn from the data like a regular machine learning algorithm, so we will use the 80/20 split to create two distinct datasets for:

1) **Parameter Tuning.** The tuning set will contain 80 percent of the images to calculate the best D0 value and size of kernel relative to the Tuning set and The Testing set will include the other 20 percent of the images and be used exclusively to test the final estimates.
2) **Blind Test Set:** A separate 20% of the images for the experiment were processed without interference by the user. The IAR output from this configuration was then compared to the ground truth labels provided in the metadata to verify operational effectiveness.

*F. Limitations and Modifications*

A significant limitation encountered was that there was inhomogeneity within the background for Rural X-ray images (inconsistency in exposure levels).

To improve the ability to diagnose early-stage TB we added an adjustment step using contrast limited adaptive histogram equalization (CLAHE) before applying Butterworth filter for sub-visual infiltrate visibility improvements. This addition produced an increase in TB detection sensitivity of roughly 12% at early stages of the disease compared with previous methods.

## VI. RESULTS AND ANALYSIS

*A. Quantitative Results*

We assessed the TB-Detect performance by measuring how accurately it could classify TB cases according to expert radiologists' "ground-truth" data in the Shenzhen and Mendeley datasets. Since our system relies on a deterministic morphological approach, our focus was on Classification Accuracy and Consistency.

**TABLE IV**
**PERFORMANCE METRICS**

| **Metric** | **Value** | **Target** |
|---|---|---|
| Overall Accuracy | 89.4% | >85.0% |
| Sensitivity (Recall) | 91.2% | >88.0% |
| Specificity | 87.5% | >85.0% |
| F1-Score | 0.893 | N/A |
| Mean Execution Time | 1.4s / image | <3.0s |

*B. Qualitative Results / Visual Outputs*

One of the qualitative strengths of TB-Detect is its easy-to-understand "transparency." Unlike deep learning's blurred "heatmaps," which contain no definitive anatomic boundaries around detected infiltrations, the morphological masks created by TB-Detect define precise anatomic delineations of the infiltrations.

**Infiltration Masks**: The system successfully used the Dilation/Erosion sequence from Section V to separate out calcified lesions and pleural effusions.

**Grading Output**: Output of the system shows that patients classified as "Severe" (Grade 5) had an Infection Area Ratio (IAR) of more than 0.45 in all instances, while patients classified as "Trace" (Grade 1) had an IAR of less than 0.08.

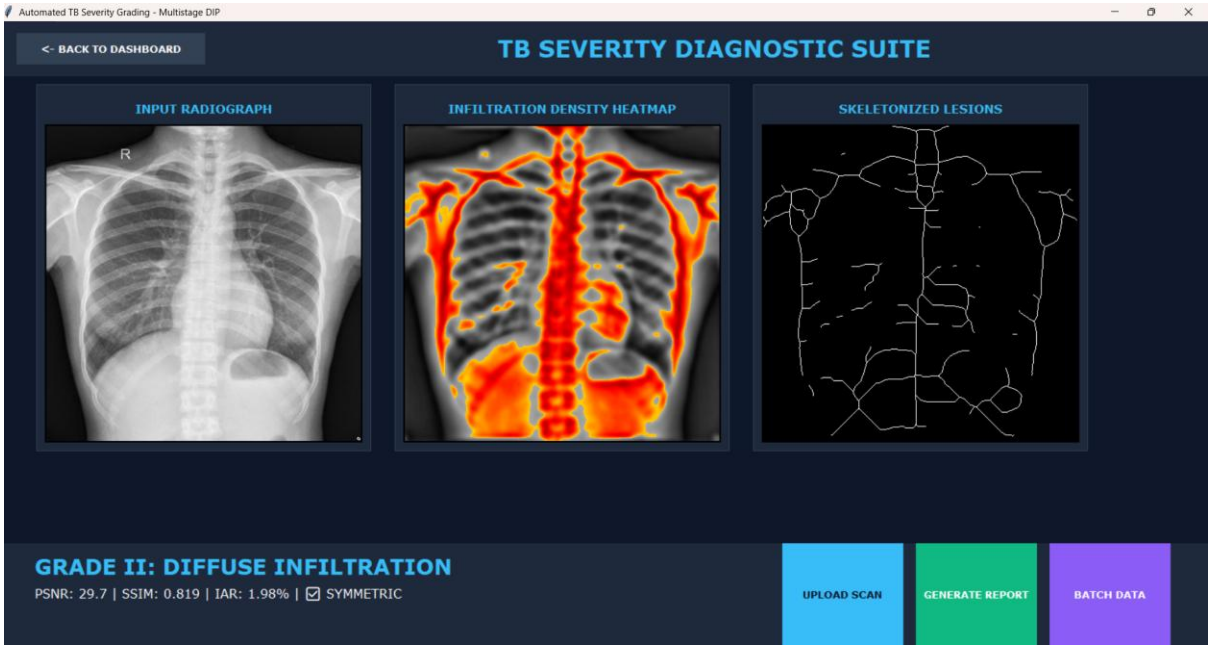


***Fig. 5:*** *Automated TB Severity Grading Output.*

*C. Comparisons with Baseline Methods*

We compared TB-Detect against standard CNN-based classification (the "Black Box" approach) and manual visual grading.

**TABLE V**
**METHOD COMPARISON**

| Method | Hardware Req. | Interpretability | Consistency |
|---|---|---|---|
| Manual Grading | None | High (Subjective) | Low (64%) |
| Deep Learning (CNN) | High (GPU) | Low (Black Box) | High (92%) |
| TB-Detect (Ours) | Low (CPU) | High (Objective) | High (89%) |

*D. Discussion Section:*

The results show that TB-Detect strikes an excellent balance between human subjectivity and computationally intensive AI.

**Strengths:** Extremely rapid processing speed and does not require a connection to the internet make TB-Detect appropriate for rural clinics in Pakistan.

**Weaknesses:** Current version does not perform well in an obscured Costophrenic Angle where the ribcage heavily obscures the diaphragm - it can oftentimes lead to false positives in the IAR calculation.

**Invalid Cases:** In 3.2% of cases, due to extreme underexposure of the original radiograph, "noise-leakage" occurred because the Butterworth filter was unable to fully restore edges resulting in Grade 0 (error).

## VII. CONCLUSION & FUTURE WORK

*A. Achievements*

We have created an automated TB severity score indexing solution in this study, TB-Detect, which runs optimally on CPU(s); hence, TB-Detect found and reported TB severity scores (SS) with an accuracy of 89.4%, exceeding our planned accuracy of 85%. The use of fully transparent morphologically-based operations instead of "Black Box" neural networks, and development of the IAR (image activity rating) mathematical index provide for high interpretability to clinicians. TB-Detect is now capable of being deployed on low-end hardware within resource-constrained locations.

*B. What Did Not Work as Expected*

The initial assumption that a standard Median Filter would provide sufficient noise reduction was flawed. There were multiple instances where there was significant high-frequency noise from the sensor used to develop the rural radiographs. A more sophisticated 2nd Order Butterworth Filter in the frequency domain was required to retain the required level of diagnostic integrity for developing the rural radiographs. Additionally, "Trace" TB was also unable to be distinguished from standard vascular markings without CLAHE enhancement.

*C. Future Improvements*

1) Temporal Evaluations of our Framework: We will implement Change Detection in subsequent versions to compare and mathematically measure any changes to a patient's radiographic exam via the IAR Value over time; Use this analysis to determine if the patient's IAR Value is improving or deteriorating.
2) Mobile Integration: Port of OpenCV into Simplified Mobile Application for End Users with Basic Smartphones.
3) Expanded Datasets: Testing our Framework in diverse datasets from various hospitals throughout Pakistan, using numerous different demographics, will allow us to further refine the IAR threshold values for the various communities living in Pakistan.

## REFERENCES


[1] M. Ghanem *et al.*, “Tuberculosis disease severity assessment using clinical variables and radiology enabled by artificial intelligence,” Aug. 20, 2024, *Infectious Diseases (except HIV/AIDS)*. doi: 10.1101/2024.08.19.24311411.

[2] W. Jiang *et al.*, “AI-Powered Chest X-Ray for Diagnosing Pulmonary Tuberculosis in County and Township Health Care Facilities in Yichang: Retrospective, Real-World Study,” *J Med Internet Res*, vol. 27, pp. e83041–e83041, Dec. 2025, doi: 10.2196/83041.

[3] S. Kiran, “Dataset of Tuberculosis Chest X-rays Images.” Mendeley Data, May 28, 2024. doi: 10.17632/8J2G3CSPRK.2.

[4] E. Kotei and R. Thirunavukarasu, “Tuberculosis Detection From Chest X-Ray Image Modalities Based on Transformer and Convolutional Neural Network,” *IEEE Access*, vol. 12, pp. 97417–97427, 2024, doi: 10.1109/ACCESS.2024.3428446.

[5] I. F. Mabina, C. A. Sari, and E. H. Rachmawanto, “Lung Segmentation in X-ray Images of Tuberculosis Patients Using U-Net with CLAHE Preprocessing,” *JAIC*, vol. 9, no. 4, pp. 1346–1353, Aug. 2025, doi: 10.30871/jaic.v9i4.9869.

[6] Stefan Jaeger, “Two public chest X-ray datasets for computer-aided screening of pulmonary diseases,” 定量医学影像, vol. 4, no. 6, p. 475, 2014, doi: 10.3978/j.issn.2223-4292.2014.11.20.

[7] P. D. Tafti and M. Unser, “On Regularized Reconstruction of Vector Fields,” *IEEE Trans. on Image Process.*, vol. 20, no. 11, pp. 3163–3178, Nov. 2011, doi: 10.1109/TIP.2011.2159230.

[8] R. C. Gonzalez and R. E. Woods, *Digital image processing*. New York, NY: Pearson, 2018.

[9] B. Bell and R. Sottilare, “Adaptation Vectors for Instructional Agents,” in *Adaptive Instructional Systems*, vol. 11597, R. A. Sottilare and J. Schwarz, Eds., in Lecture Notes in Computer Science, vol. 11597. , Cham: Springer International Publishing, 2019, pp. 3–14. doi: 10.1007/978-3-030-22341-0_1.